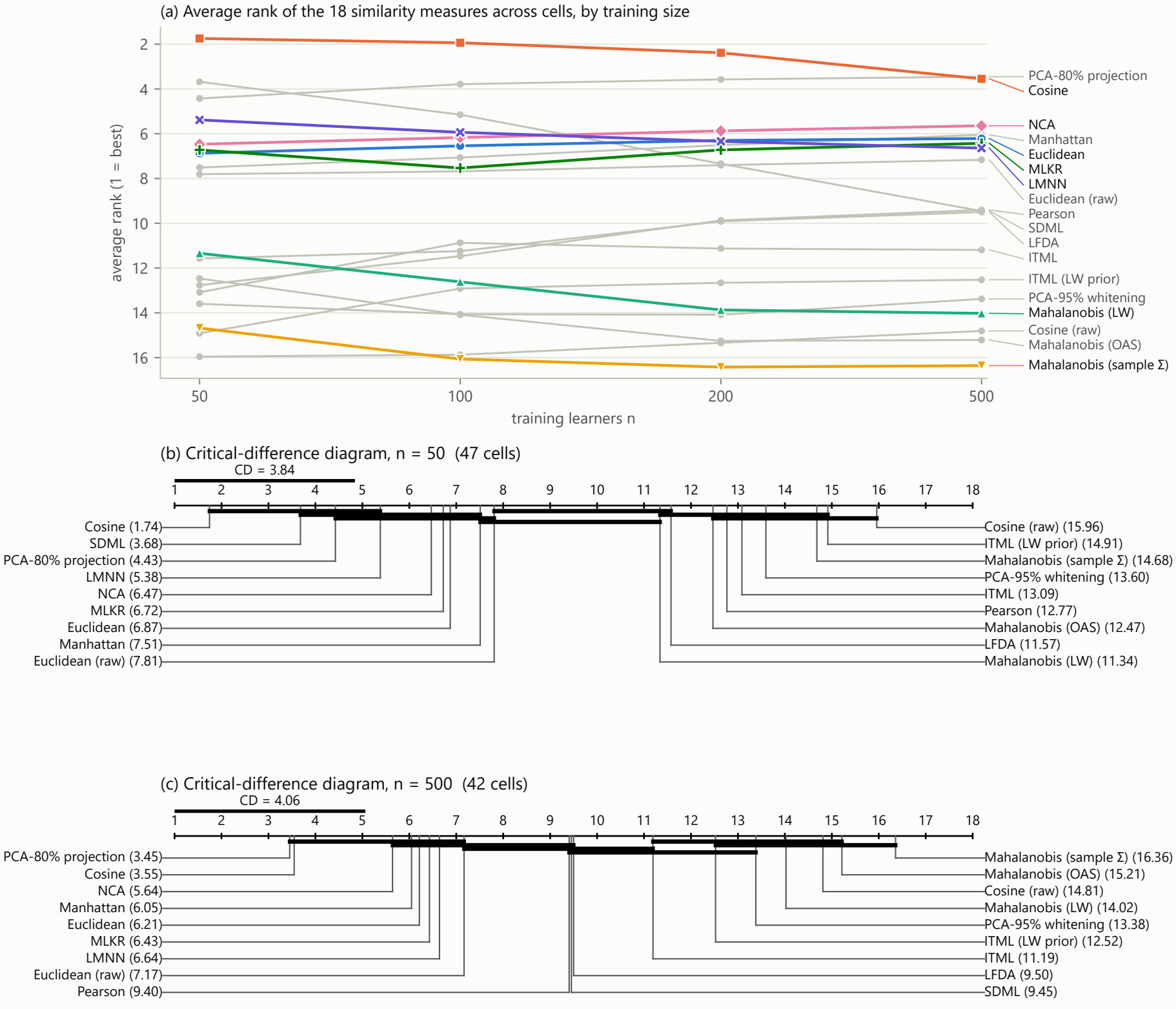
(a) Average rank of the 18 similarity measures across cells, by training size
average rank (1 = best)
training learners n
50
100
200
500
PCA-80% projection
Cosine
NCA
Manhattan
Euclidean
MLKR
LMNN
Euclidean (raw)
Pearson
SDML
LFDA
ITML
ITML (LW prior)
PCA-95% whitening
Mahalanobis (LW)
Cosine (raw)
Mahalanobis (OAS)
Mahalanobis (sample Σ)
(b) Critical-difference diagram, n = 50 (47 cells)
CD = 3.84
Cosine (1.74)
SDML (3.68)
PCA-80% projection (4.43)
LMNN (5.38)
NCA (6.47)
MLKR (6.72)
Euclidean (6.87)
Manhattan (7.51)
Euclidean (raw) (7.81)
Cosine (raw) (15.96)
ITML (LW prior) (14.91)
Mahalanobis (sample Σ) (14.68)
PCA-95% whitening (13.60)
ITML (13.09)
Pearson (12.77)
Mahalanobis (OAS) (12.47)
LFDA (11.57)
Mahalanobis (LW) (11.34)
(c) Critical-difference diagram, n = 500 (42 cells)
CD = 4.06
PCA-80% projection (3.45)
Cosine (3.55)
NCA (5.64)
Manhattan (6.05)
Euclidean (6.21)
MLKR (6.43)
LMNN (6.64)
Euclidean (raw) (7.17)
Pearson (9.40)
Mahalanobis (sample Σ) (16.36)
Mahalanobis (OAS) (15.21)
Cosine (raw) (14.81)
Mahalanobis (LW) (14.02)
PCA-95% whitening (13.38)
ITML (LW prior) (12.52)
ITML (11.19)
LFDA (9.50)
SDML (9.45)